

Question	Answer	Marks
9(a)(i)	core reduces loss of (magnetic) flux linkage/improves flux linkage	B1
9(a)(ii)	reduces (size of eddy) currents in core	B1
	(so that) heating of core is reduced	B1
9(b)	alternating voltage gives rise to changing magnetic flux in core	M1
	(changing) flux links the secondary coil	A1
	induced e.m.f. (in secondary) only when flux is changing/cut	B1

Question	Answer	Marks
10(a)(i)	penetration of beam	M1
	greater hardness means greater penetration/shorter wavelength/higher frequency/higher photon energy	A1
10(a)(ii)	greater accelerating potential difference or greater p.d. between anode and cathode	B1
10(b)	$I = I_0 \exp(-\mu x)$ ratio = $(\exp \{-1.5 \times 2.9\}) / (\exp \{-4.0 \times 0.95\}) (= \exp \{-0.55\})$	C1
	= 0.58	A1